

**|SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:25

Annexure A

Concept Mapping

Code of Conduct:

I ATHIN SIVA.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Athin Siva.S

GPS LOGISTICS TRACKING

OS CONCEPT MAP • Real-Time • Secure • Reliable

OPERATING SYSTEM

Central controller of the logistics tracking system
Manages resources, users, processes, data and communication

↓ *SUITABLE OS TYPE* ↓

REAL-TIME • MULTIUSER • NETWORK / DISTRIBUTED OS

Suitable for continuous fleet-wide location updates, low-latency processing,
high availability, security and efficient resource sharing across vehicles and servers.

↓ *OS SERVICES WORK TOGETHER* ↓

❑ USER MANAGEMENT

Authentication • permissions
Dispatcher/driver accounts • roles

⚙️ PROCESS MANAGEMENT

Scheduling • multitasking
Tracking/geofencing tasks • sync

❑ MEMORY MANAGEMENT

Allocation • virtual memory
Buffering • location data caching

❑ FILE MANAGEMENT

Create • read • write
Trip logs • access control

❑ DEVICE / I-O

Drivers • input/output
GPS receivers, sensors & modems

❑ COMMUNICATION

Sockets • IPC • networking
Vehicle/server data exchange

↕ *INTERACTION AMONG MODULES* ↕

GPS SIGNAL/
LOCATION UPDATE

SYSTEM CALL /
KERNEL

PROCESS +
MEMORY

I/O +
NETWORK DEVICE

FLEET MAP
UPDATE

↔ *RELEVANT SYSTEM CALLS* ↔

PROCESS

fork() • exec() • wait()

MEMORY

mmap() • brk() • shmget()

FILES

open() • read() • write()

NETWORK

socket() • connect() • send()

CORE GOAL → Real-time accuracy + reliability + security under heavy tracking workload

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand